

A Complete Classification of the Rational and Integral Solutions of $y^2 + z^2 = x^3 + 1$

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Abstract

We determine all rational triples (x, y, z) satisfying

$$y^2 + z^2 = x^3 + 1.$$

For a reduced fraction $x = a/b > -1$ with $b > 0$, a rational point above x exists if and only if both b and $a^3 + b^3$ are sums of two integral squares. Equivalently, every prime congruent to 3 modulo 4 occurs to an even exponent in each of these two integers. Once one representation

$$b = r^2 + s^2, \quad a^3 + b^3 = u^2 + v^2$$

is fixed, every rational point above x is obtained by multiplying the Gaussian rational $(u + iv)/(r + is)^3$ by an arbitrary norm-one element $(m + in)/(m - in)$. We prove that this parametrization is surjective. We also give a complete Gaussian-prime factorization of every integral solution, an exact formula for the number of ordered integral representations for each admissible integer x , and a factorwise refinement of the integral admissibility criterion. All required facts about sums of two squares and Gaussian integers are proved in the text.

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1 Introduction and statement of the main result

Consider the affine surface

$$\mathcal{S} : y^2 + z^2 = x^3 + 1. \quad (1.1)$$

For fixed $x \in \mathbb{Q}$, the fiber of (1.1) is a circle over $\mathbb{Q}$. The first issue is therefore not the parametrization of a circle once one rational point is known, but the arithmetic question of whether the radius squared $x^3 + 1$ is a sum of two rational squares. The correct language for both questions is the norm map

$$N : \mathbb{Q}(i) \longrightarrow \mathbb{Q}, \quad N(\alpha) = \alpha\bar{\alpha}.$$

Indeed,

$$y^2 + z^2 = N(y + iz).$$

Write

$$\Sigma_2 = \{r^2 + s^2 : r, s \in \mathbb{Z}\}.$$

We include 0 in Σ_2 . The complete answer is the following.

Theorem 1.1 (Complete rational classification). *Let $(x, y, z) \in \mathbb{Q}^3$. Then (1.1) holds precisely in the following cases.*

(i) *The boundary fiber consists of the single point*

$$(x, y, z) = (-1, 0, 0). \quad (1.2)$$

There are no rational solutions with $x < -1$.

(ii) *Suppose $x > -1$, and write $x = a/b$ in lowest terms, where*

$$a, b \in \mathbb{Z}, \quad b > 0, \quad \gcd(a, b) = 1. \quad (1.3)$$

Put

$$c = a^3 + b^3 > 0. \quad (1.4)$$

Then a rational point above x exists if and only if

$$b \in \Sigma_2 \quad \text{and} \quad c \in \Sigma_2. \quad (1.5)$$

Equivalently, for every rational prime $q \equiv 3 \pmod{4}$,

$$v_q(b) \equiv 0 \pmod{2}, \quad v_q(c) \equiv 0 \pmod{2}. \quad (1.6)$$

(iii) *Whenever (1.5) holds, choose any integers r, s, u, v satisfying*

$$r^2 + s^2 = b, \quad u^2 + v^2 = c. \quad (1.7)$$

Then every rational solution above $x = a/b$ is given by

$$\boxed{x = \frac{a}{b}, \quad y + iz = \frac{u + iv}{(r + is)^3} \frac{m + in}{m - in}}, \quad (1.8)$$

where $m, n \in \mathbb{Z}$ are not both zero. Conversely, every choice in (1.8) gives a rational solution of (1.1).

The parametrization is deliberately stated using Gaussian numbers because its structure is then transparent: the first factor in (1.8) is one point on the required circle, and the second factor runs through every rational Gaussian number of norm 1. An entirely real-coordinate version is derived in Corollary 4.1.

The proof is elementary, but completeness requires care at three points: the rational two-squares criterion, the coprimality of the numerator and denominator occurring in $x^3 + 1$, and surjectivity of the norm-one parametrization. We treat each of these explicitly.

2 Gaussian integers and sums of two squares

2.1 Euclidean and unique factorization

For $\alpha = A + iB \in \mathbb{Z}[i]$, define

$$N(\alpha) = A^2 + B^2.$$

The norm is multiplicative because

$$N(\alpha\beta) = \alpha\beta\bar{\alpha}\bar{\beta} = \alpha\bar{\alpha}\beta\bar{\beta} = N(\alpha)N(\beta).$$

Proposition 2.1. *The ring $\mathbb{Z}[i]$ is Euclidean with respect to N . Consequently it is a principal ideal domain, every pair of Gaussian integers has a greatest common divisor satisfying a Bezout identity, and $\mathbb{Z}[i]$ is a unique factorization domain.*

Proof. Let $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$. Write

$$\frac{\alpha}{\beta} = u + iv, \quad u, v \in \mathbb{R}.$$

Choose $m, n \in \mathbb{Z}$ such that

$$|u - m| \leq \frac{1}{2}, \quad |v - n| \leq \frac{1}{2},$$

and put $q = m + in$ and $r = \alpha - q\beta$. Then

$$N(r) = N(\beta) N\left(\frac{\alpha}{\beta} - q\right) = N(\beta)((u - m)^2 + (v - n)^2) \leq \frac{1}{2} N(\beta) < N(\beta).$$

Thus Euclidean division holds.

For completeness, let $I \neq (0)$ be an ideal and choose $\delta \in I \setminus \{0\}$ of minimal norm. For any $\alpha \in I$, Euclidean division gives $\alpha = q\delta + r$ with either $r = 0$ or $N(r) < N(\delta)$. Since $r \in I$, minimality forces $r = 0$. Hence $I = (\delta)$, so $\mathbb{Z}[i]$ is a principal ideal domain. In particular, the ideal generated by two elements α, β has the form (d) , and $d = A\alpha + B\beta$ for some $A, B \in \mathbb{Z}[i]$; this gives a greatest common divisor and a Bezout identity.

It remains only to justify unique factorization. In a principal ideal domain, every irreducible element π is prime. Indeed, if $\pi \mid \alpha\beta$ and $\pi \nmid \alpha$, then $\gcd(\pi, \alpha)$ is a unit, so some $A, B \in \mathbb{Z}[i]$ satisfy $A\pi + B\alpha = 1$. Multiplication by β shows that $\pi \mid \beta$. Existence of a factorization into irreducibles follows by induction on the positive integer norm: a nonzero nonunit which is not irreducible factors into two elements of strictly smaller norm. Uniqueness follows from the primality of irreducibles by the usual cancellation argument. Thus $\mathbb{Z}[i]$ is a unique factorization domain. $\square$

2.2 The obstruction at primes congruent to 3 modulo 4

We normalize the q -adic valuation by $v_q(q) = 1$.

Lemma 2.2 (Exact valuation of a sum of two squares). *Let q be a rational prime with $q \equiv 3 \pmod{4}$, and let $A, B \in \mathbb{Z}$ be not both zero. Then*

$$v_q(A^2 + B^2) = 2 \min\{v_q(A), v_q(B)\}. \quad (2.1)$$

In particular, $v_q(A^2 + B^2)$ is even.

Proof. First we show that

$$q \mid A^2 + B^2 \implies q \mid A \text{ and } q \mid B. \quad (2.2)$$

Suppose, for example, that $q \nmid B$. Then B is invertible modulo q , and

$$(AB^{-1})^2 \equiv -1 \pmod{q}.$$

This is impossible. Indeed, if $t^2 \equiv -1 \pmod{q}$, then Fermat's little theorem gives

$$1 \equiv t^{q-1} = (t^2)^{(q-1)/2} \equiv (-1)^{(q-1)/2} \equiv -1 \pmod{q},$$

because $(q-1)/2$ is odd. Thus $q \mid B$, and then $q \mid A^2$, hence $q \mid A$. This proves (2.2).

Let

$$h = \min\{v_q(A), v_q(B)\}, \quad A = q^h A_0, \quad B = q^h B_0.$$

At least one of A_0, B_0 is not divisible by q . By (2.2), this implies $q \nmid A_0^2 + B_0^2$. Therefore

$$A^2 + B^2 = q^{2h}(A_0^2 + B_0^2)$$

has valuation exactly $2h$. □

2.3 Primes congruent to 1 modulo 4

Proposition 2.3 (Fermat's theorem on primes). *Every rational prime $p \equiv 1 \pmod{4}$ is a sum of two integral squares.*

Proof. We first produce a square root of -1 modulo p . Wilson's theorem follows by pairing each nonzero residue modulo p with its inverse: all residues except 1 and -1 occur in distinct inverse pairs, so

$$(p-1)! \equiv -1 \pmod{p}.$$

Set

$$h = \left(\frac{p-1}{2}\right)!.$$

Pairing k with $p-k$ for $1 \leq k \leq (p-1)/2$ gives

$$(p-1)! \equiv (-1)^{(p-1)/2} h^2 \equiv h^2 \pmod{p},$$

because $(p-1)/2$ is even. Hence

$$h^2 \equiv -1 \pmod{p}. \quad (2.3)$$

Take a greatest common divisor

$$\delta = \gcd(p, h + i)$$

in $\mathbb{Z}[i]$, whose existence follows from Proposition 2.1. We claim that δ is not a unit. If it were a unit, a Bezout identity would give

$$Ap + B(h + i) = 1$$

for some $A, B \in \mathbb{Z}[i]$. Multiplying by $h - i$ yields

$$Ap(h - i) + B(h^2 + 1) = h - i.$$

By (2.3), both terms on the left are divisible by p in $\mathbb{Z}[i]$. Thus p would divide $h - i$, which is impossible because the imaginary coefficient of $h - i$ is -1 .

The element δ is not associated to p either, since otherwise p would divide $h + i$, again impossible from its imaginary coefficient. Since $\delta \mid p$, write $p = \delta\eta$. Both δ and η are nonunits, so

$$1 < N(\delta), N(\eta), \quad N(\delta)N(\eta) = N(p) = p^2.$$

The only possible factorization of p^2 into two integers strictly larger than 1 is $p \cdot p$. Therefore

$$N(\delta) = p.$$

Writing $\delta = A + iB$ gives $p = A^2 + B^2$. □

2.4 The integral and rational two-squares criteria

Theorem 2.4 (Integral two-squares criterion). *Let $M \geq 1$ be an integer. Then M is a sum of two integral squares if and only if every prime $q \equiv 3 \pmod{4}$ occurs to an even exponent in the prime factorization of M .*

Proof. If $M = A^2 + B^2$, Lemma 2.2 shows that $v_q(M)$ is even for every $q \equiv 3 \pmod{4}$.

Conversely, write the ordinary prime factorization as

$$M = 2^e \prod_{j=1}^r p_j^{e_j} \prod_{k=1}^s q_k^{2f_k},$$

where $p_j \equiv 1 \pmod{4}$ and $q_k \equiv 3 \pmod{4}$. By Proposition 2.3, choose $A_j, B_j \in \mathbb{Z}$ such that

$$p_j = A_j^2 + B_j^2 = N(A_j + iB_j).$$

Also $2 = N(1 + i)$ and $q_k^{2f_k} = N(q_k^{f_k})$. Hence, by multiplicativity of the norm,

$$M = N \left((1 + i)^e \prod_{j=1}^r (A_j + iB_j)^{e_j} \prod_{k=1}^s q_k^{f_k} \right).$$

The Gaussian integer inside the norm has the form $A + iB$, so $M = A^2 + B^2$. □

Theorem 2.5 (Rational two-squares criterion). *Let $R \in \mathbb{Q}$.*

- (i) *If $R < 0$, then R is not a sum of two rational squares.*
- (ii) *If $R = 0$, then its only representation is $0^2 + 0^2$.*
- (iii) *Suppose $R > 0$, and write $R = M/N$ in lowest terms with $M, N \in \mathbb{Z}_{>0}$. The following are equivalent:*

- (a) *$R = Y^2 + Z^2$ for some $Y, Z \in \mathbb{Q}$;*
- (b) *$v_q(R)$ is even for every prime $q \equiv 3 \pmod{4}$;*
- (c) *both M and N are sums of two integral squares.*

Proof. The first two assertions follow from the ordering of $\mathbb{Q}$.

Assume $R > 0$. Suppose first that $R = Y^2 + Z^2$. Choose a common denominator $D > 0$ and write $Y = A/D$, $Z = B/D$ with $A, B \in \mathbb{Z}$. Then

$$R = \frac{A^2 + B^2}{D^2}.$$

For every $q \equiv 3 \pmod{4}$, Lemma 2.2 gives

$$v_q(R) = v_q(A^2 + B^2) - 2v_q(D),$$

which is even. Thus (a) implies (b).

Now write $R = M/N$ in lowest terms and assume (b). Since $\gcd(M, N) = 1$, a prime q divides at most one of M, N . Therefore the parity of

$$v_q(R) = v_q(M) - v_q(N)$$

forces both $v_q(M)$ and $v_q(N)$ to be even. Theorem 2.4 now shows that both M and N are sums of two integral squares. Thus (b) implies (c).

Finally, assume

$$M = A^2 + B^2 = N(A + iB), \quad N = C^2 + D^2 = N(C + iD).$$

Since $N > 0$, the Gaussian integer $C + iD$ is nonzero, and

$$R = \frac{M}{N} = N \left(\frac{A + iB}{C + iD} \right).$$

Writing the quotient as $Y + iZ \in \mathbb{Q}(i)$ gives $R = Y^2 + Z^2$. Hence (c) implies (a). $\square$

3 The rational norm-one circle

The following elementary form of Hilbert's Theorem 90 is exactly what is needed to prove that the main parametrization gives every point, not merely infinitely many points.

Lemma 3.1 (All rational Gaussian numbers of norm one). *For $\eta \in \mathbb{Q}(i)$, the following are equivalent:*

(i) $N(\eta) = 1$;

(ii) there exists $\tau \in \mathbb{Q}(i)^\times$ such that

$$\eta = \frac{\tau}{\bar{\tau}}; \tag{3.1}$$

(iii) there exist integers m, n , not both zero, such that

$$\eta = \frac{m + in}{m - in}. \tag{3.2}$$

Proof. If (3.1) holds, then

$$N(\eta) = \frac{\tau}{\bar{\tau}} \frac{\bar{\tau}}{\tau} = 1.$$

Thus (ii) implies (i).

Conversely, suppose $N(\eta) = 1$. Then $\eta\bar{\eta} = 1$, so $\bar{\eta} = \eta^{-1}$. If $\eta \neq -1$, take $\tau = 1 + \eta$. This is nonzero, and

$$\bar{\tau} = 1 + \bar{\eta} = 1 + \eta^{-1} = \eta^{-1}(1 + \eta) = \eta^{-1}\tau.$$

Therefore $\tau/\bar{\tau} = \eta$. If $\eta = -1$, take $\tau = i$, for which $\tau/\bar{\tau} = i/(-i) = -1$. Hence (i) implies (ii).

Write $\tau = M + iN$ with $M, N \in \mathbb{Q}$, not both zero. Multiplying M and N by a common positive denominator does not change the quotient $\tau/\bar{\tau}$. We may therefore take $M = m$ and $N = n$ integral, proving (iii). The implication (iii) implies (ii) is immediate. $\square$

Corollary 3.2 (Classical rational parametrization of the unit circle). *Every rational pair (C, D) satisfying $C^2 + D^2 = 1$ has the form*

$$C = \frac{m^2 - n^2}{m^2 + n^2}, \quad D = \frac{2mn}{m^2 + n^2}, \quad (3.3)$$

for integers m, n not both zero, and every such pair satisfies $C^2 + D^2 = 1$.

Proof. Expand

$$\frac{m + in}{m - in} = \frac{(m + in)^2}{m^2 + n^2} = \frac{m^2 - n^2}{m^2 + n^2} + i \frac{2mn}{m^2 + n^2},$$

and apply Lemma 3.1. The identity $C^2 + D^2 = 1$ also follows directly by expanding

$$(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2.$$

□

4 Proof of the complete rational classification

Proof of Theorem 1.1. Because $y^2 + z^2 \geq 0$, every rational solution satisfies $x^3 + 1 \geq 0$, hence $x \geq -1$. If $x = -1$, then $y^2 + z^2 = 0$, so $y = z = 0$. This proves part (i).

Now suppose $x > -1$ and write $x = a/b$ as in (1.3). Then

$$x^3 + 1 = \frac{a^3 + b^3}{b^3} = \frac{c}{b^3}, \quad c = a^3 + b^3 > 0. \quad (4.1)$$

A crucial coprimality property is

$$\gcd(b, c) = 1. \quad (4.2)$$

Indeed, a common prime divisor of b and $c = a^3 + b^3$ would divide a^3 , hence a , contradicting $\gcd(a, b) = 1$. In particular, the fraction c/b^3 in (4.1) is already in lowest terms.

Assume first that a rational point (y, z) exists. By Theorem 2.5, for every prime $q \equiv 3 \pmod{4}$,

$$v_q(c) - 3v_q(b) = v_q\left(\frac{c}{b^3}\right) \equiv 0 \pmod{2}. \quad (4.3)$$

By (4.2), at most one of $v_q(b)$ and $v_q(c)$ is nonzero. If $q \mid c$, then (4.3) says directly that $v_q(c)$ is even. If $q \mid b$, it says that $3v_q(b)$ is even; since 3 is odd, $v_q(b)$ is even. Thus (1.6) holds. Theorem 2.4 gives (1.5).

Conversely, assume (1.5), and choose r, s, u, v as in (1.7). Put

$$\beta = r + is, \quad \gamma = u + iv, \quad W_0 = \frac{\gamma}{\beta^3}.$$

Since $N(\beta) = b > 0$, the denominator is nonzero. Moreover,

$$N(W_0) = \frac{N(\gamma)}{N(\beta)^3} = \frac{c}{b^3} = x^3 + 1. \quad (4.4)$$

Thus $W_0 = Y_0 + iZ_0$ supplies one rational point on the fiber.

For any integers m, n not both zero, let

$$\eta_{m,n} = \frac{m + in}{m - in}.$$

The denominator is nonzero and $N(\eta_{m,n}) = 1$. Hence

$$W = W_0 \eta_{m,n}$$

has norm $x^3 + 1$, proving that every choice in (1.8) gives a solution.

It remains to prove surjectivity. Let $W = y + iz \in \mathbb{Q}(i)$ be any Gaussian rational with

$$N(W) = x^3 + 1 = N(W_0).$$

Since $x > -1$, the common norm is positive, so $W_0 \neq 0$. Therefore

$$\eta = \frac{W}{W_0}$$

is defined and satisfies $N(\eta) = 1$. By Lemma 3.1,

$$\eta = \frac{m + in}{m - in}$$

for some integers m, n not both zero. Consequently W is exactly the value in (1.8). This proves all parts of the theorem. $\square$

Corollary 4.1 (Explicit real-coordinate parametrization). *Retain the hypotheses and notation of Theorem 1.1(ii)–(iii). Define*

$$P = r^3 - 3rs^2, \quad Q = 3r^2s - s^3. \quad (4.5)$$

Then

$$(r + is)^3 = P + iQ, \quad P^2 + Q^2 = b^3.$$

Put

$$A = uP + vQ, \quad B = vP - uQ. \quad (4.6)$$

All rational solutions above $x = a/b$ are

$$\boxed{\begin{aligned} x &= \frac{a}{b}, \\ y &= \frac{(m^2 - n^2)A - 2mnB}{b^3(m^2 + n^2)}, \\ z &= \frac{2mnA + (m^2 - n^2)B}{b^3(m^2 + n^2)}, \end{aligned}} \quad (4.7)$$

where $m, n \in \mathbb{Z}$ are not both zero.

Proof. Since $P^2 + Q^2 = N((r + is)^3) = b^3$,

$$\frac{u + iv}{(r + is)^3} = \frac{(u + iv)(P - iQ)}{b^3} = \frac{A + iB}{b^3}.$$

By Corollary 3.2,

$$\frac{m + in}{m - in} = \frac{(m^2 - n^2) + 2mni}{m^2 + n^2}.$$

Multiplying these two expressions and equating real and imaginary parts gives (4.7). Surjectivity is inherited from Theorem 1.1. $\square$

Corollary 4.2 (Exact set of rational x -coordinates). *The image of $\mathcal{S}(\mathbb{Q})$ under projection to the x -coordinate is*

$$\{-1\} \cup \left\{ \frac{a}{b} \in \mathbb{Q} : \begin{array}{l} b > 0, \gcd(a, b) = 1, a > -b, \\ v_q(b) \equiv v_q(a^3 + b^3) \equiv 0 \pmod{2} \\ \text{for every prime } q \equiv 3 \pmod{4} \end{array} \right\}. \quad (4.8)$$

For every admissible $x > -1$, the corresponding fiber is in bijection, noncanonically, with the rational unit circle.

Proof. The description of the x -coordinates is exactly Theorem 1.1. Once a base point W_0 is chosen, multiplication by W_0 gives a bijection from the norm-one elements of $\mathbb{Q}(i)$ to the Gaussian rationals of norm $x^3 + 1$. Corollary 3.2 identifies the former with the rational unit circle. $\square$

Remark 4.3 (Redundancy of the parameters). The parametrization is complete but intentionally not injective. Multiplying (m, n) by any nonzero integer leaves the quotient $(m + in)/(m - in)$ unchanged. Different choices of the representations of b and c merely choose a different base point on the same fiber; Lemma 3.1 then shows that the resulting family is still the entire fiber.

5 Complete classification of the integral solutions

We now classify all triples in $\mathbb{Z}^3$. We first record the factorization of rational primes in the Gaussian integers.

Lemma 5.1 (Rational primes in the Gaussian integers). *The following statements hold.*

(i) *The units of $\mathbb{Z}[i]$ are $\{1, -1, i, -i\}$.*

(ii) *The element $1 + i$ is a Gaussian prime and*

$$2 = -i(1 + i)^2. \quad (5.1)$$

(iii) *If $p \equiv 1 \pmod{4}$ is prime and $p = c^2 + d^2$, then*

$$p = (c + id)(c - id), \quad (5.2)$$

and the two factors are nonassociate Gaussian primes.

(iv) *Every rational prime $q \equiv 3 \pmod{4}$ remains prime in $\mathbb{Z}[i]$.*

Proof. A Gaussian integer is a unit if and only if its norm is 1, which gives (i). Since $N(1 + i) = 2$ is a rational prime, any factorization of $1 + i$ would induce a factorization of 2 into two positive integer norms; one factor must have norm 1. Thus $1 + i$ is irreducible, hence prime by Proposition 2.1. The identity (5.1) follows from $(1 + i)^2 = 2i$.

For (iii), Proposition 2.3 supplies c, d . The identity (5.2) is immediate. Each factor has norm p , so each is irreducible and hence prime. They cannot be associates. Indeed, if $c + id = \varepsilon(c - id)$ for a unit ε , comparison of real and imaginary parts forces either $c = 0$, $d = 0$, or $|c| = |d|$. The first two alternatives would make the odd prime p a square, while the last would give $p = 2c^2$, all impossible.

Finally, let $q \equiv 3 \pmod{4}$. If $q = \alpha\beta$ with both factors nonunits, then

$$q^2 = N(q) = N(\alpha)N(\beta), \quad N(\alpha), N(\beta) > 1.$$

Thus $N(\alpha) = N(\beta) = q$. This would express q as a sum of two squares, contradicting Lemma 2.2 or Theorem 2.4. Hence q is irreducible and therefore prime. $\square$

Theorem 5.2 (All integral solutions). *The integral solutions of (1.1) are classified as follows.*

(i) *For $x = -1$, the only solution is $(-1, 0, 0)$. There are no solutions for $x < -1$.*

(ii) *Let $x \geq 0$ and put*

$$M = x^3 + 1.$$

An integral pair (y, z) exists if and only if every prime $q \equiv 3 \pmod{4}$ occurs to an even exponent in M .

Assume this condition and write

$$M = 2^{e_0} \prod_{j=1}^r p_j^{e_j} \prod_{k=1}^s q_k^{2f_k}, \quad (5.3)$$

where the p_j are distinct primes congruent to 1 modulo 4 and the q_k are distinct primes congruent to 3 modulo 4. For each j , choose

$$p_j = c_j^2 + d_j^2, \quad \pi_j = c_j + id_j.$$

Then every integral pair (y, z) with $y^2 + z^2 = M$ is given uniquely by

$$\boxed{y + iz = \varepsilon(1 + i)^{e_0} \left(\prod_{k=1}^s q_k^{f_k} \right) \left(\prod_{j=1}^r \pi_j^{h_j} \overline{\pi_j}^{e_j - h_j} \right)}, \quad (5.4)$$

where

$$\varepsilon \in \{1, -1, i, -i\}, \quad 0 \leq h_j \leq e_j. \quad (5.5)$$

Here uniqueness means uniqueness of ε and of all exponents h_j after the Gaussian primes π_j have been fixed.

(iii) *The exact number of ordered integral pairs (y, z) above an admissible $x \geq 0$ is*

$$\boxed{r_2(x^3 + 1) = 4 \prod_{j=1}^r (e_j + 1)}. \quad (5.6)$$

Proof. The first assertion follows exactly as in Theorem 1.1. For $x \geq 0$, the existence criterion is Theorem 2.4.

Assume the criterion holds, and let $\alpha = y + iz \in \mathbb{Z}[i]$ satisfy $N(\alpha) = M$. By unique factorization, every Gaussian prime ρ dividing α also divides

$$\alpha \bar{\alpha} = M.$$

Writing M as the product of the rational prime powers in (5.3) and using that ρ is prime, we see that ρ divides at least one rational prime divisor of M . Lemma 5.1 factors each such rational prime in $\mathbb{Z}[i]$ and therefore lists all possible associate classes of ρ .

Let the exponent of $1 + i$ in α be t . Since $N(1 + i) = 2$, comparison of norms gives $t = e_0$. For a prime $q_k \equiv 3 \pmod{4}$, the Gaussian prime q_k has norm q_k^2 . If its exponent in α is g_k , then norm comparison gives

$$2g_k = 2f_k, \quad g_k = f_k.$$

For a split prime $p_j = \pi_j \bar{\pi}_j$, let the exponents of π_j and $\bar{\pi}_j$ in α be h_j and h'_j . Since both have norm p_j , comparison of norms gives

$$h_j + h'_j = e_j.$$

Thus $h'_j = e_j - h_j$ with $0 \leq h_j \leq e_j$. The remaining factor has norm 1 and is therefore a unit ε . This proves that every representation has the form (5.4).

Conversely, the norm of the right side of (5.4) is

$$2^{e_0} \prod_{k=1}^s q_k^{2f_k} \prod_{j=1}^r p_j^{h_j + e_j - h_j} = M,$$

so every displayed Gaussian integer yields a valid pair (y, z) . Uniqueness follows from unique factorization and from the fact that the primes π_j and $\bar{\pi}_j$ are nonassociate.

There are four choices for the unit ε and $e_j + 1$ choices for each h_j . Distinct choices yield distinct nonzero Gaussian integers by uniqueness of factorization. Hence the number of ordered pairs is (5.6). $\square$

5.1 A factorwise criterion for integral x-coordinates

The condition on $x^3 + 1$ can be refined using

$$x^3 + 1 = (x + 1)(x^2 - x + 1). \quad (5.7)$$

Proposition 5.3 (Factorwise integral admissibility). *Let $x \geq 0$ be an integer. Then $x^3 + 1$ is a sum of two integral squares if and only if all of the following hold.*

- (i) *For every prime $q \equiv 3 \pmod{4}$ with $q \neq 3$, the exponent $v_q(x + 1)$ is even.*
- (ii) *For every prime $q \equiv 3 \pmod{4}$ with $q \neq 3$, the exponent $v_q(x^2 - x + 1)$ is even. Any such prime actually satisfies $q \equiv 7 \pmod{12}$.*
- (iii) *If $3 \mid x + 1$, then $v_3(x + 1)$ is odd. If $3 \nmid x + 1$, no extra condition at 3 is needed.*

Proof. First,

$$\gcd(x + 1, x^2 - x + 1) = \gcd(x + 1, 3), \quad (5.8)$$

for reducing the second factor modulo $x + 1$ means substituting $x \equiv -1$, which gives 3. Thus a prime $q \neq 3$ divides at most one of the two factors in (5.7). Consequently, for every $q \equiv 3 \pmod{4}$, $q \neq 3$, the valuation of the product is even if and only if the valuation in the unique factor it divides is even. This proves the equivalence of the conditions away from 3.

Suppose additionally that $q \neq 3$ divides $x^2 - x + 1$. Then

$$(x + 1)(x^2 - x + 1) = x^3 + 1 \equiv 0 \pmod{q}.$$

The congruence $x \equiv -1 \pmod{q}$ is impossible, because it would give $x^2 - x + 1 \equiv 3 \pmod{q}$ and hence $q = 3$. Therefore $x^3 \equiv -1 \pmod{q}$ while $x \not\equiv -1 \pmod{q}$. The multiplicative order of x modulo q is consequently exactly 6: it divides 6, it does not divide 3 because $x^3 \equiv -1$, and it does not equal 2 because $x \not\equiv -1$. Hence $6 \mid q - 1$. Combining $q \equiv 1 \pmod{6}$ with $q \equiv 3 \pmod{4}$ gives $q \equiv 7 \pmod{12}$.

It remains to analyze $q = 3$. If $3 \nmid x + 1$, then $x \equiv 0$ or $1 \pmod{3}$, so $x^3 + 1 \not\equiv 0 \pmod{3}$ and $v_3(x^3 + 1) = 0$. If $3 \mid x + 1$, put

$$t = v_3(x + 1) \geq 1, \quad x = -1 + 3^t u, \quad 3 \nmid u.$$

A direct expansion gives

$$x^2 - x + 1 = 3 \left(1 - 3^t u + 3^{2t-1} u^2 \right).$$

The expression in parentheses is congruent to 1 modulo 3, so

$$v_3(x^2 - x + 1) = 1.$$

Therefore

$$v_3(x^3 + 1) = v_3(x + 1) + v_3(x^2 - x + 1) = t + 1,$$

which is even exactly when t is odd. Theorem 2.4 now proves the proposition. $\square$

6 Examples and infinite families

Example 6.1 (The boundary point). For $x = -1$, the right side of (1.1) is 0, and the only rational or integral point is

$$(-1, 0, 0).$$

No point exists for $x < -1$.

Example 6.2 (A negative rational x). Take

$$x = -\frac{3}{4}, \quad a = -3, \quad b = 4, \quad c = a^3 + b^3 = 37.$$

We may choose

$$4 = 2^2 + 0^2, \quad 37 = 6^2 + 1^2.$$

With $r = 2$, $s = 0$, $u = 6$, $v = 1$, and $m = 1$, $n = 0$, formula (1.8) gives

$$y + iz = \frac{6 + i}{8},$$

so

$$(x, y, z) = \left(-\frac{3}{4}, \frac{3}{4}, \frac{1}{8} \right).$$

Indeed,

$$\left(\frac{3}{4} \right)^2 + \left(\frac{1}{8} \right)^2 = \frac{37}{64} = \left(-\frac{3}{4} \right)^3 + 1.$$

Example 6.3 (A denominator obstruction). Let $x = 1/3$. Then

$$x^3 + 1 = \frac{28}{27}.$$

The prime $3 \equiv 3 \pmod{4}$ has valuation

$$v_3\left(\frac{28}{27}\right) = -3,$$

which is odd. Theorem 2.5 therefore proves that no rational y, z can satisfy (1.1) for $x = 1/3$.

Example 6.4 (An inadmissible integer). For $x = 3$,

$$x^3 + 1 = 28 = 2^2 \cdot 7.$$

The prime $7 \equiv 3 \pmod{4}$ occurs to an odd exponent, so there are no rational or integral y, z above $x = 3$.

Example 6.5 (An admissible nonsquare integer). For $x = 11$,

$$x^3 + 1 = 1332 = 2^2 \cdot 3^2 \cdot 37.$$

All primes congruent to 3 modulo 4 occur to even exponent. One representation is

$$1332 = 6^2 + 36^2,$$

so $(11, 6, 36)$ is an integral solution. Since $37 \equiv 1 \pmod{4}$ occurs to exponent 1, formula (5.6) shows that there are exactly

$$4(1 + 1) = 8$$

ordered integral pairs (y, z) above $x = 11$.

Proposition 6.6 (An infinite polynomial family). *For every $t \in \mathbb{Q}$,*

$$(x, y, z) = (t^2, 1, t^3) \tag{6.1}$$

is a rational solution. More generally, for every $t \in \mathbb{Q}$ and all integers m, n not both zero,

$$\boxed{\begin{aligned} x &= t^2, \\ y &= \frac{m^2 - n^2 - 2mnt^3}{m^2 + n^2}, \\ z &= \frac{2mn + (m^2 - n^2)t^3}{m^2 + n^2} \end{aligned}} \tag{6.2}$$

is a rational solution. In particular, taking $t \in \mathbb{Z}$ in (6.1) gives infinitely many integral solutions.

Proof. The identity

$$1^2 + (t^3)^2 = t^6 + 1 = (t^2)^3 + 1$$

proves (6.1). Multiplying $1 + it^3$ by the norm-one number

$$\frac{m + in}{m - in} = \frac{m^2 - n^2 + 2mni}{m^2 + n^2}$$

and equating real and imaginary parts yields (6.2). Since multiplication by a norm-one element preserves the norm, every displayed triple is a solution. $\square$

7 Conclusion

The rational and integral solution sets of (1.1) are governed exactly by the norm map from $\mathbb{Q}(i)$. For a reduced rational $x = a/b > -1$, the two independent arithmetic obstructions are that the denominator b and the coprime numerator $a^3 + b^3$ of $x^3 + 1$ must each be integral sums of two squares. Once these obstructions vanish, a single Gaussian-rational point generates the entire rational fiber by multiplication with norm-one elements. In the integral case, unique factorization in $\mathbb{Z}[i]$ not only decides existence but lists every solution and counts them exactly.